

Constant Elasticity of Substitution (CES) function for CGE Model

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Constant elasticity of substitution (CES) Production function

- General form of CES production function:

$$Q = \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}}$$

- X_i = the quantity of input i
 Q = the quantity of output
 δ_i = share parameter for input i and $-1 < \rho < \infty$ is parameter
- With technical change terms:

$$Q = \frac{1}{A} \left\{ \sum_i \delta_i \left(\frac{X_i}{A_i} \right)^{-\rho} \right\}^{-\frac{1}{\rho}}$$

Elasticity of substitution

- Let's define σ :

$$\sigma = \frac{1}{1 + \rho}, 0 < \sigma < \infty$$

- We can substitute ρ with σ :

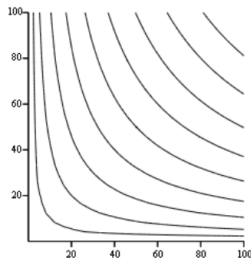
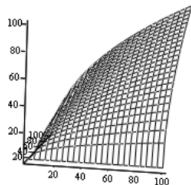
$$Q = \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}} = \left\{ \sum_i \delta_i X_i^{-\frac{1-\sigma}{\sigma}} \right\}^{-\frac{1}{\frac{1-\sigma}{\sigma}}}$$
$$Q = \left\{ \sum_i \delta_i X_i^{\frac{\sigma-1}{\sigma}} \right\}^{\frac{\sigma}{\sigma-1}}$$

CES production function with varying parameters

- A two variables K , capital and L , labor:

$$Q = \left(\delta_K K^{\frac{\sigma-1}{\sigma}} + \delta_L L^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$$

- Surface and contour plot with $\delta_K = 0.5$, $\delta_L = 0.5$ and $\sigma = 0.9$



More about Sigma

- As you learned from the exercise, σ measure the curvature of an isoquant of the CES production function.
- σ is called the elasticity of substitution, it measures:
 - How easy it is for a producer to substitute between or among factors of production as responses to changing relative factor prices
 - How the change in the slope of the isoquant affect the factor intensity (ratio of two factors of production)
 - The percentage change in factor intensity as the effect of 1 percent change in factor prices ratio or marginal rate of technical substitution.
- $\sigma = 0.5$ means, when factor price ratio, let's say w/r increase by 1 percent, factor intensity will increase by 0.5.

Special cases of CES production function

- Cobb-Douglas Production function - A CES with $\sigma = 1$

$$Q = K^\alpha L^\beta \text{ or more generally: } Q = \prod_i X_i^{\alpha_i}$$

- Leontief production function - A CES with $\sigma = 0$

$$Q = \min \left(\frac{K}{a}, \frac{L}{b} \right) \text{ or more generally: } Q = \min \left(\frac{X_1}{A_1}, \dots, \frac{X_n}{A_n} \right)$$

- Linear (perfect complement) production function - A CES with $\sigma = \infty$

$$Q = mK + nL, \text{ or more generally: } Q = \sum_i a_i X_i$$

Cost minimization

- A producer's problem is to minimize cost of production subject to a given production technology, or:

$$\min_{X_i} \sum_i W_i X_i \text{ s.t. } Q = \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}}$$

where W_i is the price of input i .

- To solve, we use the Lagrange function:

$$L = \sum_i W_i X_i + \lambda \left\{ Q - \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}} \right\}$$

- and find the first order conditions:

$$\frac{\partial L}{\partial X_i} = 0 \text{ for all } i = 1, \dots, n \text{ and } \frac{\partial L}{\partial \lambda} = 0$$

Demand for or optimum quantity of factors of production

- From the first order conditions we can derive (See derivation in the handout, and ask if you get lost):

$$X_i = Q \delta_i^\sigma \left(\frac{W_i}{c} \right)^{-\sigma}$$

where:

$$c = \left(\sum \delta_i^\sigma W_i^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$$

and σ is the elasticity of substitution, and $0 < \sigma < \infty$.

- c is the unit cost or average cost of production or:

$$c = \frac{C}{Q} = \frac{\sum_i W_i X_i}{Q}$$

See the derivation in the handout.

- Query: *why is the average cost a constant? shouldn't it be U-shaped?*

Demand for factors and unit cost

- You will see many of this in the equations of CGE models:

$$X_i = Q \delta_i^\sigma \left(\frac{W_i}{c} \right)^{-\sigma}$$

$$c = \left(\sum \delta_i^\sigma W_i^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$$

- Make yourself familiar (if not memorize), and think about what is the effect of changing Q and W_i into X_i .
 - X_i increase proportionally with Q , other things held constant: homotheticity and constant returns to scale
 - X_i decrease with W_i (its own price), other things held constant
 - X_i increase with W_j (other input's price), other things held constant

Factor intensity and factor price ratio

- Factor intensity is the ratio of two optimal quantity of factors of production:

$$\frac{X_i}{X_j} = \frac{Q \delta_i^\sigma \left(\frac{W_i}{c}\right)^{-\sigma}}{Q \delta_j^\sigma \left(\frac{W_j}{c}\right)^{-\sigma}} = \left(\frac{\delta_i}{\delta_j} \frac{W_j}{W_i}\right)^\sigma$$

- Factor intensity, $\frac{X_i}{X_j}$, is a function of factor price ratio, $\frac{W_j}{W_i}$. This is a feature of a homothetic production function.
- This relationship explains the meaning of elasticity of substitution σ .
 - $\sigma = 1.5$ means that when factor price ratio $\frac{W_j}{W_i}$, increase by 1, factor intensity, $\frac{X_i}{X_j}$, will increase by 1.5.

Illustration with 2 factors production function

- With production function $Q = (\delta_K K^{-\rho} + \delta_L L^{-\rho})^{-\frac{1}{\rho}}$, and $\sigma = \frac{1}{1+\rho}$, the demand for factors of production are:

$$K = Q \delta_K^{\sigma} \left(\frac{r}{c} \right)^{-\sigma}$$

$$L = Q \delta_L^{\sigma} \left(\frac{w}{c} \right)^{-\sigma}$$

where

$$c = (\delta_K^{\sigma} r^{1-\sigma} + \delta_L^{\sigma} w^{1-\sigma})^{\frac{1}{1-\sigma}}$$

- For the purpose of visualization, the following equations define:
 - Isoquant: $K^Q(Q, L) = \left(\frac{Q^{\frac{\sigma-1}{\sigma}} - \delta_L L^{\frac{\sigma-1}{\sigma}}}{\delta_K} \right)^{-\frac{\sigma}{1-\sigma}}$
 - Isocost curve: $K^C(Q) = \frac{cQ - wL}{r}$
 - Factor intensity: $f = \left(\frac{\delta_K}{\delta_L} \frac{w}{r} \right)^{\sigma}$

- The demand for inputs with CES technology, derived from cost minimization:

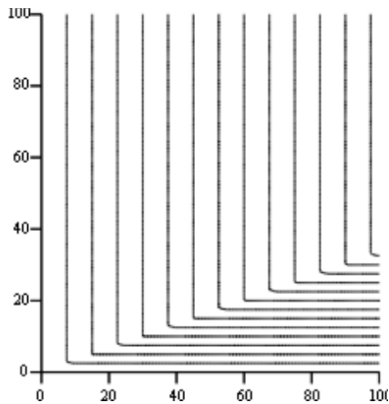
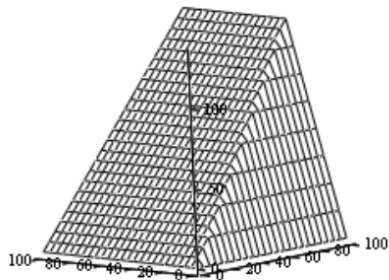
$$X_i = Q \delta_i^\sigma \left(\frac{W_i}{c} \right)^{-\sigma}$$

where

$$c = \left(\sum \delta_i^\sigma W_i^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$$

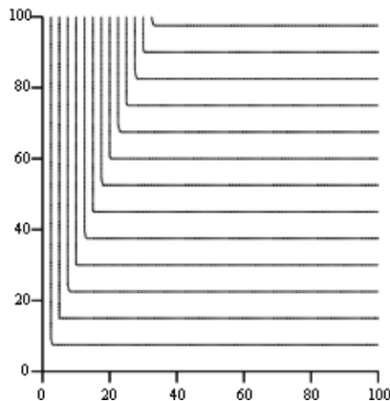
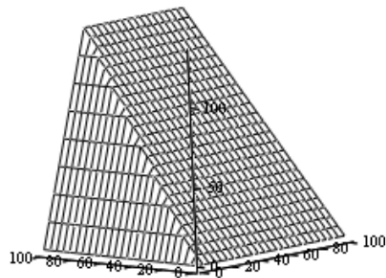
Leontief Production Function ($\sigma = 0$) – Curvature

$$Q = \min \left\{ \frac{K}{A_K}, \frac{L}{A_L} \right\}, \text{ with } A_K = 0.25, A_L = 0.75$$



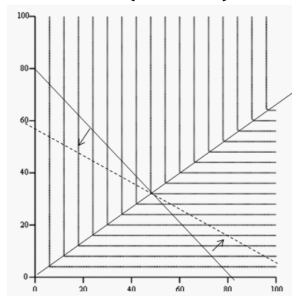
Leontief Production Function - Curvature

$$Q = \min \left\{ \frac{K}{A_K}, \frac{L}{A_L} \right\}, \text{ with } A_K = 0.75, A_L = 0.25$$



Leontief production function is input-price independent

- $Q = \min \left\{ \frac{K}{A_K}, \frac{L}{A_L} \right\}$, with $A_K = 0.4, A_L = 0.6$



- Factor intensity $f = \frac{K}{L} = \frac{0.4}{0.6} = \frac{2}{3}$ independent of prices, or $\sigma = 0$
- Cost minimizing inputs: $K = 0.4Q, L = 0.6Q$

Nested Production function

- Suppose we have a 3-inputs production function:

$$Q = f(U, S, K)$$

where: Q = Output, U = Unskilled labor
 S = Skilled labor, and K = Capital

- The producer's standard problem is to:

$$\min w_U U + w_S S + rK \text{ subject to } Q = f(U, S, K)$$

where w_U , w_S and r are unskilled-wage, skilled-wage, and return to capital respectively.

- The standard procedure for optimization can be used to find the cost-minimizing U , S and K :

$$L = w_U U + w_S S + rK + \lambda(Q - f(U, S, K))$$

Nested production function

- Now, suppose the production function looks like this:

$$Q = \left(\delta_L (\delta_U U^{-\rho_L} + \delta_S S^{-\rho})^{\frac{\rho}{\rho_L}} + \delta_K^{-\rho} \right)^{-\frac{1}{\rho}}$$

- This is an example of the *nested*, or *separable* production function, because we can re-write:

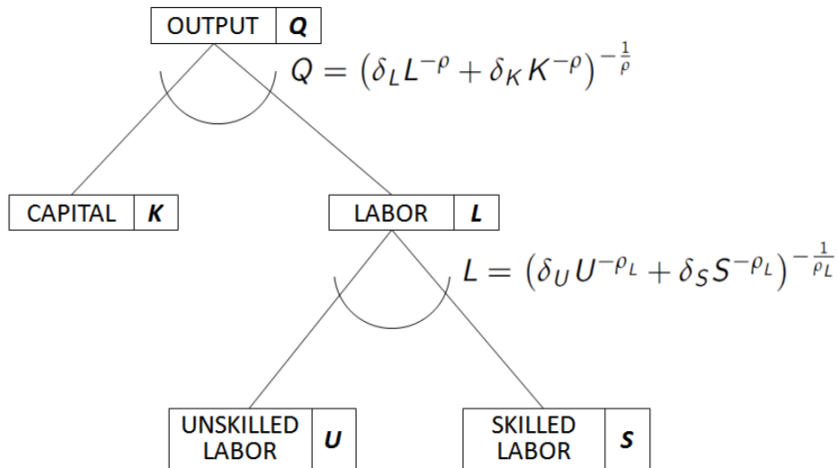
$$Q = f(L, K) = (\delta_L L^{-\rho} + \delta_K K^{-\rho})^{-\frac{1}{\rho}}$$

where

$$L = g(U, S) = (\delta_U U^{-\rho_L} + \delta_S S^{-\rho_L})^{-\frac{1}{\rho_L}}$$

- L is often called a composite variable because it is a composite or CES aggregation of U and S .
- Both $f(\cdot)$ and $g(\cdot)$ have CES form but with different parameter, thus different elasticity of substitution.

Diagram of a nested production function



Cost minimization with nested production function

- How to solve the minimization problem with nested production function like this?

$$\min_{U,S,K} w_U U + w_S S + rK \text{ s.t. } Q = \left(\delta_L (\delta_U U^{-\rho_L} + \delta_S S^{-\rho_L})^{\frac{\rho}{\rho_L}} + \delta_K^{-\rho} \right)^{-\frac{1}{\rho}}$$

- We can do it in steps, by dividing the problem into two problems:
- Problem #1: Find optimum combination of U and S for a given L (composite labor):

$$\min_{U,S} w_U U + w_S S \text{ s.t. } L = (\delta_U U^{-\rho_L} + \delta_S S^{-\rho_L})^{-\frac{1}{\rho_L}}$$

- Problem #2: Find optimum combination of K and L for a given Q :

$$\min_{K,L} wL + rK \text{ s.t. } Q = (\delta_L L^{-\rho} + \delta_K K^{-\rho})^{-\frac{1}{\rho}}$$

where w is the price of composite labor L .

Solution to problem #1

- Problem:

$$\min_{U,S} w_U U + w_S S \text{ s.t } L = \left(\delta_U U^{-\rho_L} + \delta_S S^{-\rho_L} \right)^{-\frac{1}{\rho_L}}$$

- We can apply the standard formula:

$$X_j = Q \delta_j^{\frac{1}{1+\rho}} \left(\frac{w_j}{c} \right)^{-\frac{1}{1+\rho}} \text{ and } c = \left(\sum \delta_i^{\frac{1}{1+\rho}} w_i^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}} \text{ to find:}$$

$$U = L \cdot \delta_U^{\frac{1}{1+\rho_L}} \left(\frac{w_U}{w} \right)^{-\frac{1}{1+\rho_L}} \quad (1)$$

$$S = L \cdot \delta_S^{\frac{1}{1+\rho}} \left(\frac{w_S}{w} \right)^{-\frac{1}{1+\rho_L}} \quad (2)$$

$$w = \left(\delta_U^{\frac{1}{1+\rho_L}} w_U^{\frac{\rho_L}{1+\rho_L}} + \delta_S^{\frac{1}{1+\rho_L}} w_S^{\frac{\rho_L}{1+\rho_L}} \right)^{\frac{1+\rho_L}{\rho_L}} \quad (3)$$

Solution to problem #2

- Problem:

$$\min_{L,K} wL + rK \text{ s.t } Q = (\delta_L L^{-\rho} + \delta_K K^{-\rho})^{-\frac{1}{\rho}}$$

- We can apply the standard formula:

$$X_j = Q \delta_j^{\frac{1}{1+\rho}} \left(\frac{W_j}{c} \right)^{-\frac{1}{1+\rho}} \text{ and } c = \left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}} \text{ to find:}$$

$$L = Q \cdot \delta_L^{\frac{1}{1+\rho}} \left(\frac{w}{c} \right)^{-\frac{1}{1+\rho}} \quad (4)$$

$$K = Q \cdot \delta_K^{\frac{1}{1+\rho}} \left(\frac{r}{w} \right)^{-\frac{1}{1+\rho}} \quad (5)$$

$$c = \left(\delta_L^{\frac{1}{1+\rho}} w^{\frac{\rho}{1+\rho}} + \delta_K^{\frac{1}{1+\rho}} r^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}} \quad (6)$$

Complete solution

Equation 1 to 9 define the solution of the problems: the optimum quantity of U, S, K for a given Q, w_U, w_S, r :

$$\textcircled{1} \quad w = \left(\delta_U^{\frac{1}{1+\rho_L}} w_U^{\frac{\rho_L}{1+\rho_L}} + \delta_S^{\frac{1}{1+\rho_L}} w_S^{\frac{\rho_L}{1+\rho_L}} \right)^{\frac{1+\rho_L}{\rho_L}}$$

$$\textcircled{2} \quad c = \left(\delta_L^{\frac{1}{1+\rho}} w^{\frac{\rho}{1+\rho}} + \delta_K^{\frac{1}{1+\rho}} r^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}}$$

$$\textcircled{3} \quad K = Q \cdot \delta_K^{\frac{1}{1+\rho}} \left(\frac{r}{w} \right)^{-\frac{1}{1-\rho}}$$

$$\textcircled{4} \quad L = Q \cdot \delta_L^{\frac{1}{1+\rho}} \left(\frac{w}{c} \right)^{-\frac{1}{1+\rho}}$$

$$\textcircled{5} \quad U = L \cdot \delta_U^{\frac{1}{1+\rho_L}} \left(\frac{w_U}{w} \right)^{-\frac{1}{1+\rho_L}}$$

$$\textcircled{6} \quad S = L \cdot \delta_S^{\frac{1}{1+\rho}} \left(\frac{w_S}{w} \right)^{-\frac{1}{1+\rho_L}}$$

Solution to problem #2

- Problem:

$$\min_{L,K} wL + rK \text{ s.t } Q = (\delta_L L^{-\rho} + \delta_K K^{-\rho})^{-\frac{1}{\rho}}$$

- We can apply the standard formula:

$$X_j = Q \delta_j^{\frac{1}{1+\rho}} \left(\frac{W_j}{c} \right)^{-\frac{1}{1+\rho}} \text{ and } c = \left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}} \text{ to find:}$$

$$L = Q \cdot \delta_L^{\frac{1}{1+\rho}} \left(\frac{w}{c} \right)^{-\frac{1}{1+\rho}} \quad (7)$$

$$K = Q \cdot \delta_K^{\frac{1}{1+\rho}} \left(\frac{r}{w} \right)^{-\frac{1}{1+\rho}} \quad (8)$$

$$c = \left(\delta_L^{\frac{1}{1+\rho}} w^{\frac{\rho}{1+\rho}} + \delta_K^{\frac{1}{1+\rho}} r^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}} \quad (9)$$

Complete solution

Equation 1 to 9 define the solution of the problems: the optimum quantity of U, S, K for a given Q, w_U, w_S, r :

$$\textcircled{1} \quad w = \left(\delta_U^{\frac{1}{1+\rho_L}} w_U^{\frac{\rho_L}{1+\rho_L}} + \delta_S^{\frac{1}{1+\rho_L}} w_S^{\frac{\rho_L}{1+\rho_L}} \right)^{\frac{1+\rho_L}{\rho_L}}$$

$$\textcircled{2} \quad c = \left(\delta_L^{\frac{1}{1+\rho}} w^{\frac{\rho}{1+\rho}} + \delta_K^{\frac{1}{1+\rho}} r^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}}$$

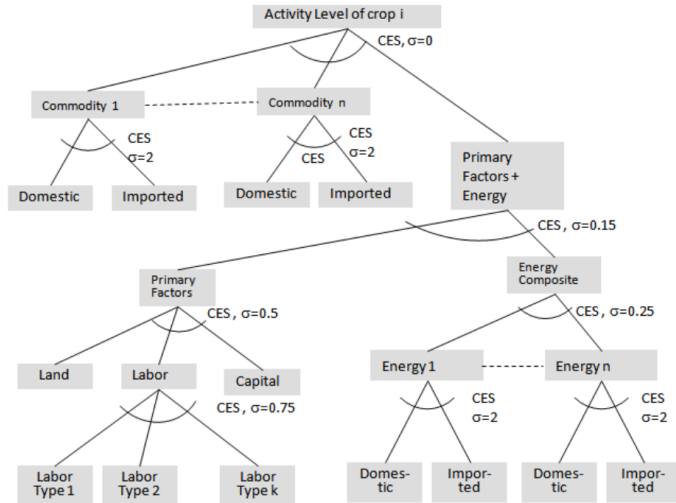
$$\textcircled{3} \quad K = Q \cdot \delta_K^{\frac{1}{1+\rho}} \left(\frac{r}{w} \right)^{-\frac{1}{1-\rho}}$$

$$\textcircled{4} \quad L = Q \cdot \delta_L^{\frac{1}{1+\rho}} \left(\frac{w}{c} \right)^{-\frac{1}{1+\rho}}$$

$$\textcircled{5} \quad U = L \cdot \delta_U^{\frac{1}{1+\rho_L}} \left(\frac{w_U}{w} \right)^{-\frac{1}{1+\rho_L}}$$

$$\textcircled{6} \quad S = L \cdot \delta_S^{\frac{1}{1+\rho}} \left(\frac{w_S}{w} \right)^{-\frac{1}{1-\rho_L}}$$

Nesting can be complex, but standard principle is the same



Assignment #2

Write down the equations that define the solution to the following problem:

$$\text{Minimize cost: } C = w_U U + w_S S + p_N N + rK + p_E E + p_D X_D + p_M X_M$$

$$\text{Subject to: } Q = \min \left\{ \frac{f_\rho (f_{\rho_{US}} (U, S), f_{\rho_{KE}} (K, E), N)}{a_1}, \frac{f_{\rho_{DM}} (X_D, X_M)}{a_2} \right\}$$

Where:

U = unskilled labor

w_U = unskilled wage

S = skilled labor

w_S = skilled wage

N = land

p_N = return to land

E = energy

p_E = price of energy

K = capita

r = return to capital

X_D = domestic raw material

p_D = price of X_D

X_M = imported raw material

p_M = price of X_M

$f_\rho, f_{\rho_{US}}, f_{\rho_{KE}}, f_{\rho_{DM}}$ is a CES function with parameter $\rho, \rho_{US}, \rho_{KE}, \rho_{DM}$ respectively.

a_1 and a_2 is the Leontief technical coefficient.

Assignment #2: Diagram

